

Saadiyah Liakhat Ali [Profile](#)

UAE Golden Visa Holder

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University of Sharjah

SUMMARY

Dynamic and results-driven chemistry enthusiast with a Bachelor of Science (Hons.) degree in Chemistry, graduating with Highest Honors from the University of Sharjah. Completed Master of Science in Chemistry and worked as a Graduate Research Assistant at University of Sharjah. Also serving as President of the American Chemical Society International Student Chapter at the esteemed University of Sharjah, showcasing strong leadership abilities and exceptional communication skills. Eager to embark on a rewarding career with forward thinking organization that values and leverages my skills in analysis, technology, and computation. Passionate about research and innovation, I am poised to make a meaningful impact in academia.

EDUCATION

MSc in Chemistry	University of Sharjah
College of Sciences	2024 – Present
Thesis Supervisor: Dr. Tanveer ul Haq	4.00 CGPA

BSc in Chemistry (Highest Honors)	University of Sharjah
College of Sciences	2020 – 2024
Senior Project: Surface Engineering of Self-supported Electrodes for Sustainable Hydrogen Production from Sea water.	3.90 CGPA
Theoretical Title: Fe-Based Ru Replacement Anode: Advancing Green Hydrogen Production from Sea water.	
Supervisor: Dr. Tanveer ul Haq	

Higher Secondary Education	New Indian Model School, Dubai
Final Project: Chemical Determination of Soft Drinks	2016 – 2019
Supervisor: Ms. Sindu Samad	96.2 %

WORK EXPERIENCE & PROFESSIONAL QUALIFICATIONS

President of American Chemical Society Student Chapter	Sharjah, UAE
University of Sharjah	2025-Present

Graduate Research Assistant	Sharjah, UAE
University of Sharjah	2024 - Present

President of Chemistry Students Association	Sharjah, UAE
University of Sharjah	2024 - 2025
Description: ACS Conference Speaker at NYU, Abu Dhabi.	
Representing UOS at various platforms	

Peer Tutor and Advisor	Sharjah, UAE
University of Sharjah	2023 - 2024
Description: Employed by Peer Tutoring / Advising Office, Student Success Hub, DASS, University of Sharjah.	
Certified for providing quality tutoring for students;	
Recognized for outstanding dedication and commitment to the enrichment of peers through the service.	

General Department of Forensic Science and Criminology, Dubai Police HQ

Summer Training Intern, 7 weeks on-site

Description: Exploring the disciplines of Forensic Science, training on dealing with forensic evidence and chemical analysis using spectroscopic techniques.

Dubai, UAE

Summer 2023

Undergraduate Research Assistant

University of Sharjah

Research Project: Synthesis and DFT calculations of fluorescent dansyl compounds for latent fingerprints visualization.

Supervisor: Dr. Mahreen Arooj

Sharjah, UAE

2021 - 2022

PUBLICATIONS

1. Coupling Electronic Modulation with Oxyanion Chemistry for Chloride-Resistant Seawater Electrolysis. *Chemical Communications*, 2026 (First Author). DOI: [10.1039/D6CC02785G](https://doi.org/10.1039/D6CC02785G).
2. Resolving the Activity–Stability Paradox in Seawater Oxygen Evolution via Boron-Stabilized, Fluorine-Modified NiFe Heterostructures (First-author article; under review in *ChemSusChem*).
3. Interface-Engineered Electrode Materials for Stable Alkaline Seawater Splitting: Insights into Failures and Alternatives (First-authored article; under review in *Next Research*).
4. Integrating AI Tools in Chemistry Education: Pedagogical Applications of DeepLigBuilder and Chemistry42 (First-author article; accepted in the Scopus-indexed book for the proceeding of “*The Second Sharjah International Conference on Education, AI and Beyond*”).
5. Unravelling the Language of Blood - A Comprehensive Study on Bloodstain Pattern Analysis (Author: Saadiyah)- (Al Noor, T. & Al-Obaidi, Omar. (2024). ACS Regional MEA Conference -UAE) {Conference Proceedings in Research Gate}.

OTHER RESEARCH PROJECTS

MarsSustainaCell

University of Sharjah

2024

The project presents an Aluminum-CO₂/MOF Battery, that works on account of Carbon-Capture, thereby, providing electrical energy output from atmospheric carbon dioxide emissions via chemical energy to electrical energy conversion. This can be used for both intrinsic applications and extraterrestrial exploration activities like Mars mission to power the MMRTG system.

Nuclear Battery / Propulsion for Mars mission

University of Sharjah

2023

The project addressed the absence of a cost-feasible and corrosion resistant radiation shielding technology in Multi-mission Radioisotope Thermoelectric Generator for Mars mission and proposed Metal-Organic-Frameworks as effective shielding mitigators that absorb harmful radiation that help minimize radiation exposure risks to both astronauts and spacecraft components.

A New Insect-Attracting agent to revolutionize sustainable agriculture: Propyl Butyrate

And its Application on the UAE national plant – Calotropis Procera

University of Sharjah

2022

The project presented an innovative technology to safeguard the UAE national plant - Calotropis Procera.

Piezoelectric tires to illuminate the future of Automobile Industry

2022

University of Sharjah

A New Technique for visualization of Latent Fingerprints using Spices and Condiments

2021

University of Sharjah

The project provided a safe, cost-effective, and innovative technique to visualize latent fingerprints and thus contributed to the field of Forensic Chemistry.

SKILLS

Public Speaking and Communication: Proficient Speaking skills

Presentation skills: Effective delivery skills

Project Management: Organizing and managing projects, timelines, and resources efficiently.

Research Skills: Ability to gather and analyze information, critically evaluate sources, and synthesize findings.

Analytical Skills: Analyzing data, identifying patterns, and drawing conclusions.

Problem-Solving: Identifying problems or challenges and developing innovative solutions.

Technical Skills: Laboratory instrumentation (IR, Raman, NMR, XPS, XRD, ICP, GC-MS etc.), spectroscopic methods.

AWARDS AND HONORS

10th Sharjah Chamber Award for Innovators – 2nd Prize for my research project on a nicotine-capturing device integrated with AI-driven real-time exposure monitoring to combat youth addiction. **Cash Prize: 15,000 AED.** 2026

NextGen Innovators Competition (Organized by The Arab-German Young Academy of Sciences and Humanities (AGYA)) – 2nd Prize for Research Project on Green Hydrogen production 2025

ACS Student Communities Engagement Award – 1000 USD granted to host regional UAE-ACS events.

ACS Starter Grant – 350 USD granted to support student activities in the campus.

Design for Future Higher Education Competition – 2nd prize in the competition held at University of Sharjah.

Project Title: AI-Powered Sensory Control System for Neurodiverse Students

UAE Innovates Award from Chancellor Awarded by Prof. Humaid al-Naimiy, former Chancellor of University of Sharjah for my contribution to innovation with “**MarsSustainaCell**” and excellent academic performance.

Inter-University Regional Student Research Nexus 2.0 (Organized by Amity University)- 1st prize for best Research Project. 2024

8th Sharjah Chamber Award for Innovators – 1st prize for my project titled: “**MarsSustainaCell**” against 87 other outstanding projects. The competition was organized by UOS in collaboration with Sharjah Chamber of Commerce and Industry and welcomed participants from all universities across UAE. **Cash Prize: 20,000 AED.**

FANR Student Hackathon Award – 2nd place in the student hackathon organized by Federal Authority for Nuclear Regulations, UAE for my project “**Nuclear Battery for Mars mission**” – Road to COP28. 2023

Sharjah Chamber Award for Innovators – 3rd prize for my project: “**A New Technique for visualization of Latent Fingerprints using Spices and Condiments**”. **Prize: 10,000 AED [Featured in top UAE Newspapers]** 2021

Noble Prize Competition at UOS - Winner, received a **symbolic award** and certificate from Dean of College of Sciences. 2020

Chemistry Festival – 2nd prize winner for the project: “**A Sustainable Route to Fingerprinting**”. 2023

Annual Open Day at UOS Awarded for representing the Chemistry Department 2024

Annual Scientific Student Societies Exhibition Recognized for serving as the president of chemistry club. 2024

Chemistry Festival – 2nd prize winner for the project: “**Application of Computational Chemistry in Life**”. 2022

Chemistry Festival – 3rd place for the project: “**Boosting Immune system during Covid-19**”. 2021

Youth Sustainability Competition Award – Certificate of recognition for being among the **top 10** best projects 2021

INVITED ACADEMIC AND PUBLIC TALKS

The 12th Chemistry Festival: The Impact of AI on Chemistry University of Sharjah
Invited to deliver a presentation based on my Master Thesis. 2026

American Chemical Society Regional Meeting American University of Sharjah
Represented the ACS–UOS International Student Chapter; presented its mission, vision, and activities. 2025

American Chemical Society Conference New York University
Selected as a speaker at the prestigious ACS Conference held in NYU, Abu Dhabi. 2024
Delivered a presentation on Blood Stain Pattern Analysis.

Industrial Training University of Sharjah
Chosen to deliver a talk/presentation based on my 2023 Summer Training at Dubai Police. 2023

RAK Excellence Awards	Sharjah, UAE
Post-graduate poster competition held at American University of Ras Al Khaimah	2026
UAE Graduate Students Research Conference	Sharjah, UAE
Research conference held at University of Sharjah.	2026
Design For Future Higher Education Competition	Sharjah, UAE
2nd prize in the competition held at University of Sharjah. Cash Prize: 2,000 AED	2025
Project: “Luma: AI-Powered Sensory-Inclusive Learning Environment for Neurodiverse Students”	
Design For Future Higher Education Competition	Sharjah, UAE
3rd prize in the competition held at University of Sharjah. Cash Prize: 1,000 AED	2024
Project: “Blockchain-powered Credentialing for Education”	
President of Chemistry Club	Sharjah, UAE
I was elected, by the committee, as the president of the chemistry club at University of Sharjah.	2023 – 2024
Innobator Competition	2022
Selected as the top 5 projects in the competition sponsored by Innovation Box LLC.	
Youth Sustainability Competition	2022
Certified to be among the best 10 projects out of 88 group projects in the competition sponsored by Bee’ah.	
Student Councilor	Sharjah, UAE
Certified for serving as a student councillor at the council committee of UOS.	2021 – 2022